Weather-Based Crop Advisory System

Software Requirements Specification

Academic Project Documentation

1. Purpose

This Software Requirements Specification defines what the Weather-Based Crop Advisory System should do, the users who interact with it, the data involved, external interfaces and the quality requirements expected from the system.

2. Scope

The application provides weather-based decision support to farmers. It uses a selected location, crop and optional crop-stage information, retrieves weather information and evaluates configured rules to produce crop-specific advisory messages.

3. Definitions and Abbreviations

Term	Meaning
API	Application Programming Interface used to communicate with an external service.
Advisory	A recommendation generated from weather and crop-related conditions.
Crop Stage	The current growth stage of a selected crop.
Weather Service	External service providing weather observations and/or forecasts.
Admin	Authorized user who manages application data and rules.

4. Overall Description

A farmer logs in, selects a location and crop, and views weather information. The backend obtains weather data from the external weather service. The advisory engine compares the data with active rules and returns relevant recommendations. Administrators manage crop records and advisory rules.

4.1 User Classes

User	Description	Permissions
Farmer	Normal user seeking crop and weather guidance.	Login, profile, location, crop, weather, advisory and alerts.
Administrator	Maintains system data and advisory configuration.	Manage crops, crop stages and advisory rules.

5. System Features

5.1 Authentication

• User registration and login.

• Protected access to authenticated features.

• Role-based access for administrator functions.

5.2 Location Management

• Select or enter a farming location.

• Use the selected location for weather retrieval.

• Show an error if the location cannot be resolved.

5.3 Crop Management

• Select a supported crop.

• Provide crop stage when required.

• Admin can add, update or deactivate crop entries.

5.4 Weather Retrieval

• Request current and forecast weather for the selected location.

• Display useful values such as temperature, precipitation/rain information, humidity and wind where available.

• Handle weather-service errors without producing unsupported recommendations.

5.5 Advisory Engine

The advisory engine evaluates weather conditions against predefined rules. A rule can contain a crop or crop stage, a condition, a recommendation message and an optional severity. Only active and applicable rules are evaluated.

Condition Example	Possible Advisory
High probability of rain	Consider postponing irrigation when immediate watering is not required.
Strong wind	Avoid spraying during unsuitable wind conditions.
Very high temperature	Check soil moisture and irrigation needs more frequently.
High humidity with wet conditions	Increase attention to weather-related crop risk.

5.6 Advisory Display

• Show recommendation messages in a readable form.

• Group or label advisories by type/severity where practical.

• Include the relevant weather condition when possible.

5.7 Alerts

• Display important weather-related warnings.

• Do not create an alert when required weather data is missing.

5.8 Administration

• Manage crop information.

• Manage crop-stage information.

• Create, update, activate or deactivate advisory rules.

6. External Interface Requirements

User Interface: The interface should provide login/registration, farmer dashboard, location and crop forms, weather cards, advisory messages and admin screens. The layout should remain usable on common screen sizes.

Weather Interface: Communication with the selected weather provider should use HTTP/HTTPS. Provider-specific code should be isolated so another provider can be used later if necessary.

7. Data Requirements

Entity	Important Data
User	User ID, name, email/login data, password hash and role.
Location	Location ID, name and coordinates/location identifier.
Crop	Crop ID, name, supported stages and status.
WeatherData	Location, timestamp, temperature, precipitation/rain, humidity, wind and forecast values.
AdvisoryRule	Rule ID, crop/crop stage, condition, message, severity and status.
Advisory	Generated message, rule reference, user reference and timestamp.

8. Functional Requirement Specifications

ID	Specification
SRS-F01	The system shall authenticate registered users before protected features are used.
SRS-F02	The system shall store or use the selected location for weather retrieval.
SRS-F03	The system shall retrieve weather data for the selected location.
SRS-F04	The system shall allow selection of a supported crop.
SRS-F05	The system shall evaluate applicable rules against retrieved weather data.
SRS-F06	The system shall generate crop-specific advisory messages when rule conditions are satisfied.
SRS-F07	The system shall display weather information and advisories on the dashboard.
SRS-F08	The system shall restrict administrative operations to authorized administrators.
SRS-F09	The system shall handle weather-service failure gracefully.
SRS-F10	The system shall allow administrators to manage active advisory rules.

9. Non-Functional Requirements

ID	Requirement
NFR-01	Usability: common farmer tasks should be understandable without technical knowledge.
NFR-02	Performance: normal dashboard operations should respond within reasonable time for expected usage.
NFR-03	Security: passwords should be securely hashed and protected routes should use authorization.
NFR-04	Reliability: missing weather data must not be treated as a valid condition.
NFR-05	Maintainability: weather integration and advisory logic should be modular.
NFR-06	Portability: the application should be deployable in a standard development/server environment.
NFR-07	Scalability: new crops and rules should be addable without changing the complete architecture.

10. Business Rules

1. Only active advisory rules are considered.

2. A rule is applicable only when its configured crop and weather conditions match.

3. Missing weather values must not be assumed to satisfy a condition.

4. An advisory is guidance and must not be presented as a guaranteed result.

5. Changes to rules apply to future advisory generation after activation.

11. Use Case Summary

Use Case	Actor	Goal
Register	Farmer	Create an account.
Login	Farmer/Admin	Access the correct dashboard.
Set Location	Farmer	Choose the location used for weather data.
Select Crop	Farmer	Choose the crop for advisory generation.
View Weather	Farmer	View current and forecast weather.
Get Advisory	Farmer	Receive crop-specific recommendations.
View Alerts	Farmer	See important weather warnings.
Manage Crops	Admin	Maintain crop records.
Manage Rules	Admin	Maintain advisory conditions and messages.

12. Error Handling

• Invalid login → show an authentication error.

• Invalid location → request a valid location.

• Weather service unavailable → show that weather data is temporarily unavailable.

• Incomplete weather data → skip rules requiring missing fields.

• Unauthorized admin request → reject the request.

13. Security Requirements

• Passwords must not be stored as plain text.

• Authentication state must be protected.

• Role-based authorization must protect administrative operations.

• External API credentials must not be exposed in client-side code or committed directly to a public repository.

14. Future Enhancements

• Regional and multilingual advisories.

• SMS/push notifications.

• More crops and crop stages.

• Historical weather and advisory records.

• Soil and irrigation information.

• Sensor/IoT integration.

15. Conclusion

The SRS defines a practical, limited-scope decision-support system that connects weather information with crop-specific recommendations. Its modular structure keeps the implementation manageable while providing a base for future improvements.